

Calabi–Yau Quantum Groups

The Companion

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Abstract

A Calabi–Yau category has a trace; a chiral algebra has an operator product; no formal argument connects them. Volume III constructs the connection as a two-stage, dimension-indexed correspondence $\Phi_d = \mathrm{Sp}_{\Sigma_{d-1}, C}^{\mathrm{ch}} \circ \Phi_d^{\mathrm{FA}}$: a canonical E_d -holomorphic factorization algebra on the Calabi–Yau target, then factorization homology along a $(d-1)$ -cycle restricted to a reference curve. Inside the first stage lives the quantum group: the critical cohomological Hall algebra $Y^+(X)$ whose Drinfeld double $G(X)$ is the level-three object; at the terminal scalar level every $K_3 \times E$ reading lands on the Igusa form, $Z_{\mathrm{DT}}^{\mathrm{red}} = -4096 \Delta_5^{-2}$. This companion states the functor with its honest per-dimension status, the κ -spectrum that stratifies the landscape, the proved scalar/automorphic layer, the no-go theorems that fence the conditional algebraic layer, and the finite recognition data that separate the two.

I The seed

A single Calabi–Yau input carries a spectrum of numerical shadows, one per construction class. At $X = K_3 \times E$ the five measurements

$$\kappa_{\mathrm{cat}}(X) = 0, \quad \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}(X) = 0, \quad \kappa_{\mathrm{ch}}^{\mathrm{Heis}}(X) = 3, \quad \kappa_{\mathrm{BKM}}(\Delta_5) = 5, \quad \kappa_{\mathrm{fiber}}(K_3) = 24$$

are all correct and mutually incommensurable: the Künneth holomorphic Euler characteristic, the Hodge supertrace, the rank of the Heisenberg–Cartan after Gram projection, the Borcherds weight $c_1(0)/2$ of the Gritsenko denominator, and the Mukai rank of the fibre. No additive relation holds among them -- the once-tempting identity $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}}^{\mathrm{Hodge}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ predicts 2 where the value is 5. The object of the volume is the mechanism producing all five from one datum, and the mechanism is a functor in two stages,

$$\Phi_d^{(\Sigma_{d-1}, C)} = \mathrm{Sp}_{\Sigma_{d-1}, C}^{\mathrm{ch}} \circ \Phi_d^{\mathrm{FA}} : \mathrm{CY}_d\text{-Cat} \xrightarrow{\Phi_d^{\mathrm{FA}}} E_d\text{-HolFA}(X) \xrightarrow{\mathrm{Sp}_{\Sigma_{d-1}, C}^{\mathrm{ch}}} E_{n(d)}\text{-ChirAlg}(C),$$

with native operadic level $n(1) = \infty$, $n(2) = 2$, $n(d \geq 3) = 1$. Stage one assembles a holomorphic factorization algebra on the target from the degree- $(1-d)$ Gerstenhaber bracket on Hochschild cohomology, by Kontsevich–Tamarkin formality -- pinned up to a $\mathrm{GRT}_1(\mathbb{Q})$ -torsor -- together with Costello–Gwilliam locality and the holomorphic BV twist of the volume form. Stage two is specialisation, never inversion: different cycles produce different chiral shadows of one stage-one datum. At $d \geq 3$ the output is natively E_1 , because $\mathrm{Conf}_2(\mathbb{R}^3)$ is simply connected; the braiding that makes a quantum

group lives one level up, on the Drinfeld centre of the representation category, where the half-braiding on evaluation modules is the spectral R -matrix $R(z) = \sigma_{V_u}(V_v)$, $z = u - v$.

2 The objects

Definition 2.1 (Calabi–Yau category). *A CY_d -category is a smooth proper pretriangulated dg (or cyclic A_∞) category $\mathcal{A}$ with a non-degenerate negative-cyclic trace $\mathrm{Tr}_{\mathcal{A}}^-: HC_d^-(\mathcal{A}) \rightarrow k$; its Hochschild trace is only the decategorified shadow. Models: $\mathrm{Perf}(X)$ with the Serre trace for projective Calabi–Yau X ; matrix-factorisation and critical-CoHA categories for Landau–Ginzburg charts. The holomorphic volume form Ω_X rigidifies the trace into the BV datum of stage one.*

Definition 2.2 (The quantum group). *Inside the stage-one object lies the positive effective moduli stack with its Kontsevich–Soibelman critical structure; its equivariant vanishing-cycle cohomology is the positive half $Y^+(X) = H_{\mathrm{eq}}^\bullet(\mathcal{M}_{\mathrm{eff}}^+(X), \phi_W)$ with the Schiffmann–Vasserot Hall product. When the Hall pairing (from Serre duality), Cartan, completion, integral form (Davison–Meinhardt) and stable-envelope transport (Maulik–Okounkov) are constructed, the Drinfeld double $G(X) = Y^+ \bowtie Y^\circ \bowtie Y^-$ is the quantum group of the datum. The equivariance stratifies by geometry: full torus on toric charts; reduced class with residual $\mathbb{C}^\times$ on compact targets; orbifold inertia; period-domain equivariance on lattice-polarised families.*

Definition 2.3 (The K_3 column). $\Phi_2(D^b \mathrm{Coh}(K_3)) = \mathcal{H}_{\mathrm{Muk}}$, the Mukai–Heisenberg chiral algebra: twenty-four currents $J_i(z)$ with $J_i J_j \sim \omega_{\mathrm{Muk}}^{ij}/(z-w)^2$ on the Mukai lattice $\Pi_{4,20} = U^4 \oplus E_8(-1)^2$, character $1/\eta^{24}$, $\kappa_{\mathrm{ch}}^{\mathrm{Hodge}} = 2$. The $d = 2$ bulk is the K_3 Yangian, the double of the K_3 cohomological Hall algebra on $\bigsqcup_n \mathrm{Hilb}^n(K_3)$; the Mukai-form-preserving envelopes form a trichotomy -- ungraded orthogonal $Y_h(\mathfrak{so}(4, 20))$, the Hodge-parity super-extension $Y_h(\mathfrak{so}(4|20))$ (non-Kac: the Mukai pairing is symmetric on both parities), with Kac $\mathfrak{osp}(4|20)$ excluded. The trigonometric lift is twenty-four coupled copies of quantum toroidal $\mathfrak{gl}_1$, with Miki symmetry the mapping-class $SL_2(\mathbb{Z})$ of the elliptic factor.

Definition 2.4 (The $K_3 \times E$ object). $\mathbf{H}_{\Delta_5}$ is the conditional Hall–Drinfeld double of $\mathrm{CoHA}_{K_3 \times E}$, a quasi-Hopf object of the fourth kind: the super-Etingof–Kazhdan quantisation of a Manin pair $(\mathfrak{g}_{\Delta_5}, \mathfrak{n}_+^{\mathrm{imag}} \oplus \mathfrak{h}^{\mathrm{imag}})$ -- no Manin triple exists, since the signature- $(2, 1)$ Cartan form has isotropic rank one -- twisted by the Siegel–Borchers associator built from $\Phi_{10}^{\mathrm{un}}/\eta^{24}$ and carrying a projective $\widetilde{M}_{24}$ symmetry. Its character is a theorem; its existence as an algebra is a finite recognition problem.

3 The theorems

Theorem 3.1 (Two-stage factorisation; per-dimension status). $\Phi_d = \mathrm{Sp}_{\Sigma_{d-1}, C}^{\mathrm{ch}} \circ \Phi_d^{\mathrm{FA}}$ with $n(d)$ as above. Stage one is proved unconditionally at $d \leq 2$; at $d = 3$ on the smooth proper object-level locus, granted the framing package (negative-cyclic CY class, the Costello-corrected chain-level S^3 -framing $\{b, B_{\mathrm{TCFT}}^{(2)}\} = \circ$, and $\pi_3(B\mathrm{Sp}) = \circ$); on explicit toric and formal charts beyond. Morphism-level functoriality is conjectural at every d ; the collection $\{\Phi_d\}$ is a correspondence programme, not one functor across dimensions. The claims $n(2) = 2$ and $n(1) = \infty$ are conditional enhancements (the braided enhancement needs the chain-level two-sphere framing action; the constructed $d = 1$ output has singular operator product and is not commutative).

Theorem 3.2 (The m_3 obstruction, resolved in three levels). *On a non-formal cyclic A_∞ -category of dimension three the strict identity $[m_3, B^{(2)}] = \circ$ is false -- witness $[m_3, B_{\mathrm{term}}^{(2)}][a|a|a|b] = 2\alpha[b] \neq$*

o. *What holds is the moduli-level identity $\{b, B_{\text{TCFT}}^{(2)}\} = 0$ (Costello's open-closed topological conformal field theory: strip degenerations against the genus-change node, $\partial^2 = 0$ on moduli chains), and a filtered derived comparison. The chain-level identification of the two $B^{(2)}$'s is the named open frontier of stage one at $d = 3$; three tempting wrong proofs (cyclic invariance, the mixed-complex axiom, Tsygan formality) are dissected as mathematics. On the Fermat quintic strict formality is refuted outright: the Yukawa coupling $Y_3 = 5 + 2875q + \dots$ is a nonzero m_3 , and the correct statement is curved formality with Yukawa as the BCOV curving datum.*

Theorem 3.3 (Centre and braiding). *For dualizable $\text{Rep}^{E_1}(A)$ the Drinfeld centre satisfies $\mathcal{Z}(\text{Rep}^{E_1}(A)) \simeq \text{Rep}^{E_2}(Z_{\text{ch}}^{\text{der}}(A))$ as braided monoidal ∞ -categories, and on the chiral Koszul locus $Z_{\text{ch}}^{\text{der}}(A) = C_{\text{ch}}^{\bullet}(A, A)$. On toric charts, granted the Morita-kernel packages, the three level-three constructions meet: $\text{Rep}^{E_2}(Z_{\text{ch}}^{\text{der}}(A)) \simeq \mathcal{Z}(\text{Rep}^{E_1}(A)) \simeq \mathcal{Z}(\text{Mod}(Y^+(X))) \simeq \text{Rep}^{E_2}(G(X))$ -- three faces carrying three associativity classes. The evaluation chain $\text{CoHA}(\mathbb{C}^3) = Y^+(\widehat{\mathfrak{gl}}_1) \hookrightarrow Y \xrightarrow{\text{ev}} \text{End}(\mathcal{W}_{1+\infty}[\lambda]\text{-vac})$ keeps the three arrows distinct: the cohomological Hall algebra is the positive half, never $\mathcal{W}_{1+\infty}$ itself; its character is the MacMahon function $\prod_n (1 - q^n)^{-n}$.*

Theorem 3.4 (Dimension stratification of the Hodge shadow; proved). *For the Hodge supertrace $\Xi(X) = \sum_q (-1)^q h^{o,q}(X)$ on compact Calabi-Yau d -folds: $\Xi = 0$ at every odd d (Serre pairs cancel); $\Xi = 2$ for even d with full $SU(d)$ holonomy; $\Xi = n + 1$ for hyperkähler $K_3^{[n]}$ -type. The two even strata meet only at $d = 2$. Local values: $\kappa_{\text{ch}}^{\text{loc}}(\mathbb{C}^3) = 1$, resolved conifold 1, local $\mathbb{P}^2 = 3/2$, $K_S = \chi_{\text{top}}(S)/2$; on the quintic the BV-corrected reading is $\chi_{\text{top}}/24 = -25/3$ while the Hodge reading is 0 -- distinct invariants, both correct. The scalar genus tower is $\text{obs}_g = \kappa_{\text{ch}} \lambda_g$ on the uniform-weight lane, with the first cross-channel correction $\delta F_2(\mathcal{W}_3) = (c + 204)/(16c)$ at multi-weight.*

Theorem 3.5 (Abelian K_3 Yangian; proved). *The $d = 2$ bulk admits the presentation: twenty-four currents with Mukai-diagonal double poles; Miura transfer matrix $T_{K_3}(u) = \prod_{i=1}^{24} (u - \phi_i)$; structure function $g_{K_3}(u) = \prod_{i=1}^{24} \frac{u - h_i}{u + h_i}$ with the Calabi-Yau constraint $\sum_i h_i = 0$ and unitarity $g(u)g(-u) = 1$; Drinfeld coproduct $\Delta_z(T(u)) = T(u) \otimes T(u - z)$; Fock characters $p_{24}(n) = \chi(\text{Hilb}^n K_3)$. Koszul duality acts by $h_i \mapsto -h_i$, with conductor $\kappa + \kappa' = 0$. The non-abelian extension -- the lattice-Hopf matching with vertex operators -- is a conjecture.*

Theorem 3.6 (The scalar layer at $K_3 \times E$; proved). *With Δ_5 the weight-five Gritsenko-Nikulin form, $D_5 = 64^{-1} \Delta_5$, $\Phi_{10}^{\text{un}} = \Delta_5^2$ the Igusa cusp form χ_{10} :*

$$Z_{\text{DT,red}}^{K_3 \times E} = -(\Phi_{10}^{\text{OP}})^{-1} = -4096 \Delta_5^{-2} \quad (\text{Oberdieck-Pandharipande-Pixton}),$$

$$Z_{\text{red}}^X(p, q, t) = \prod_{(n,m,\ell) > 0} (1 - p^n q^m t^\ell)^{-c(nm,\ell)} = \frac{pqt}{\Delta_5^2} \quad (\text{Katz-Klemm-Vafa via Pandharipande-Thomas}),$$

with c the coefficients of the K_3 elliptic genus $2\phi_{0,1}$ -- the exponent doubling is exactly the reciprocal-square mechanism. The rank-one sector is Göttsche's $\sum_n \chi(\text{Hilb}^n K_3) q^n = q/\eta^{24}$; the Humbert restriction of Δ_5^{-2} produces the doubled Heisenberg character η^{-48} exactly (Gritsenko-Nikulin $\Delta_5 = (2\pi i z) \eta(\tau)^{24} \eta(\sigma)^{24} (1 + O(z^2))$; and the Weyl-Kac-Borcherds denominator of $\mathfrak{g}_{\Delta_5}$ is $64^{-1} \Delta_5(2Z)$ with signed multiplicities $f(nm, l)$. The universal Borcherds weight identity reads $\kappa_{\text{BKM}}(\Phi_N) = c_N(0)/2$, always naming the input form; on the CHL ladder $N \in \{1, 2, 3, 4, 6\}$ the corrected weights are $(5, 3, 2, \frac{3}{2}, 1)$ with $c_N(0) = (10, 6, 4, 3, 2)$ (Jatkar-Sen, Govindarajan-Krishna; the once-recorded ladder $(5, 4, 3, 2, 1)$ is retracted); the Fake Monster form Φ_{12} carries $c_\Lambda(0) = 24$, $\kappa_{\text{BKM}} = 12$.

Theorem 3.7 (Six routes: falsification and stratification). *Six constructions aim at $G(K_3 \times E)$: the functor image, the Borceherds lift, the Mukai lattice vertex algebra, the Kummer orbifold, the sigma-model half-twist, and the class- $\mathcal{S}$ realisation. Unconditionally, the six-way isomorphism is false: the route algebras stratify into exactly three κ_{ch} -strata, $(3, 3, 3)$, (12) , (24) , and the comparison from the functor route to the lattice route is injective with cokernel a rank-twenty-one Heisenberg. The convergence statement that survives is a colimit: granted four chain-level hypotheses (Costello–Li factorisation for holomorphic Chern–Simons; the threefold Kummer Mayer–Vietoris halving $24 \rightarrow 12$; half-twist primitive extraction; the Hochschild–Kostant–Rosenberg–Borceherds lift), the pentagon of intertwiners has positive-half colimit $Y_{K_3 \times E}^+$, and with Hopf pairing and completion data $G(K_3 \times E) = D_h(Y^+)$. Three automorphic shadows of the convergence are proved.*

Theorem 3.8 (No-go theorems; proved). (i) *No smooth projective toric Calabi–Yau threefold exists (ray generators on an affine hyperplane contradict fan completeness).* (ii) *For compact non-toric X the identification of $\Phi_3(X)$ with the double of a quiver CoHA is obstructed: toric completeness, connected automorphisms, finite-quiver equivariance and the BCOV anomaly $\alpha_{\text{BCOV}} = \frac{\chi_{\text{top}}}{24} \text{tr At}(T_X) \in H^1(X, \mathcal{O}_X)$ interlock into two and a half independent mechanisms.* (iii) *$K_3 \times E$ admits no global noncommutative crepant resolution (five obstructions); the Serre-equivariant quasi-resolution substitute has forced character $-(\Phi_{10}^{\text{OP}})^{-1}$.* (iv) *No K_3 -fibred threefold specialisation realises the Fake Monster: $\text{rk } \Lambda_{\text{Leech}} = 24$ exceeds every attainable Picard rank; its home is conjecturally $d = 5$ on $K_3 \times K_3 \times E$.* (v) *Toric positive halves define no Borceherds weight: the κ_{BKM} -invariant requires a weak Jacobi input, an even Lorentzian lattice, and a Weyl vector.*

Theorem 3.9 (Wall-crossing as gauge equivalence; proved). *The cohomological Hall algebra is an algebra, not a coalgebra; its ordered bar complex is a separate coassociative object, and the square-zero equation lives on the Ginzburg dg Lie algebra, where the potential’s critical equation is the Maurer–Cartan equation. Kontsevich–Soibelman wall-crossing is Maurer–Cartan gauge equivalence on the motivic Hall Lie algebra; the pentagon identity is the Jacobi identity in degree three; conifold wall-crossing is cluster mutation; the attractor mechanism selects the canonical gauge representative. Shadow-tower truncations are not gauge-invariant across walls.*

Conjecture 3.10 (The three bridges). (i) *Chain fusion: the stage-two output A_X is the boundary chart $\text{End}(b)$ of a canonical vacuum in an open factorization category on a tangential log curve -- verified on $\mathbb{C}^3$, local $\mathbb{P}^2$, the conifold, conditional at $K_3 \times E$, open in general.* (ii) *CY-C: a braided Hopf datum $C(\mathcal{A}, q)$ with $\text{Rep}^{\text{fd}} \simeq$ the modular tensor category of $A_{\mathcal{A}}$, recovering Kazhdan–Lusztig at pt/G .* (iii) *Mirror equals Koszul: $\Phi(\mathcal{A}^\vee) \simeq \Phi(\mathcal{A})^\dagger$; at $d = 2$ this is false -- homological mirror symmetry preserves $\kappa_{\text{ch}} = 2$ while Koszul duality negates it -- and at $d = 3$ it is the conjecture that mirror symmetry flips the sign of χ_{top} .*

4 The computations

The κ -table, five subscripts never conflated:

X	κ_{cat}	$\kappa_{\text{ch}}^{\text{Hodge}}$	$\kappa_{\text{ch}}^{\text{Heis}}$	κ_{BKM}	κ_{fiber}
$K_3 \times E$	0	0	3	5 (at Δ_5)	24
K_3	2	2	--	--	--
$K_3^{[n]}$	$n+1$	$n+1$	--	--	--
quintic	0	0 (BV: $-25/3$)	--	--	--
local $\mathbb{P}^2$	--	$3/2$ (local)	--	--	--
conifold, $\mathbb{C}^3$	--	1 (local)	--	--	--

The K_3 trichotomy: three $c = 24$ chiral algebras with three distinct characteristics -- the functor output $\mathcal{H}_{\text{Muk}}(\kappa_{\text{ch}} = 2)$, the Monster orbifold V^h ($\kappa = 12$), the Leech lattice vertex algebra ($\kappa = 24$) -- ratio $24 : 12 : 2$; three inequivalent readings of “the” K_3 algebra.

The Δ_5 dictionary, one normalisation trap stated once: Δ_5 theta-normalised with leading coefficient 64; $D_5 = 64^{-1}\Delta_5$ monic; $\Phi_{10}^{\text{un}} = \Delta_5^2 = \chi_{10}$; $\Phi_{10}^{\text{OP}} = D_5^2 = 4096^{-1}\Delta_5^2$. The four-image Ψ -landscape of bosonic Borchers lifts terminates: Monster (non-geometric, Cartan rank two on $\Pi_{1,1}$), Fake Monster (Φ_{12} , rank twenty-six, conjecturally $d = 5$), Igusa/ K_3 (Δ_5 , rank three), Enriques; the Conway module is the super-image boundary, not a fifth row; the twenty-two non-Leech Niemeier Borchers–Kac–Moody algebras are genuine counterexamples to surjectivity.

The Vol I interface constants: the derived-centre complementarity bucket is $\{0, 8, 13, 250/3, 25/3\}$ on the canonical G/L/C/M/B witnesses (the B-row value $8 = 2c_+(\Pi_{4,20}) = \text{ord}(H_1)$, with $\hbar^2 \cdot 8 = -1$; the Bershadsky–Polyakov entry $25/3$ conditional on its anomaly-ratio lane, the once-recorded $98/3$ retracted); distinct from the $K_3 \times E$ five-tuple $(0, 0, 3, 5, 24)$ -- two different fives sharing one cell.

5 The garden

One logarithmic 1-form generates the programme. On a smooth complex curve X , the form $\eta = d \log(z_1 - z_2)$ is the unique coordinate-free 1-form with a residue along the diagonal of $\text{Conf}_2(X)$; the Arnold relation $\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0$ generates all relations among its higher companions. From η one builds, for an augmented chiral algebra $\mathcal{A}$ on X , the ordered bar complex $\overline{B}_X^{\text{ord}}(\mathcal{A}) = T^c(\mathcal{A}) \otimes_{\log}^{\bullet} (\overline{\text{FM}}_{\bullet}(X))$; its differential squares to zero exactly when $\mathcal{A}$ satisfies the Borchers identity. Assembled over $\overline{\mathcal{M}}_{g,n}$ with the prime-form propagator, the total differential is a Maurer–Cartan element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$; genus ≥ 1 flatness fails by the curvature $d_{\text{fib}}^2 = \kappa(\mathcal{A}) \omega_g$, and the scalar $\kappa \dashv \kappa(\text{Vir}_c) = c/2$, $\kappa(V_k(\mathfrak{g})) = \dim \mathfrak{g} (k + h^\vee)/2h^\vee$ -- is the central charge read as modular curvature. The Verdier–Koszul dual obeys conductor laws $c(\mathcal{A}) + c(\mathcal{A}^\dagger) = K$ with $K(\text{Vir}) = 26$, $K(\mathcal{W}_N) = 4N^3 - 2N - 2$, $K(V_k(\mathfrak{g})) = 2 \dim \mathfrak{g}$ along $k \mapsto -k - 2h^\vee$; the landscape stratifies into archetype classes G/L/C/M of shadow depth $2/3/4/\infty$ plus the critical Feigin–Frenkel stratum.

Every object sits on one chain: primitive factorization category $\rightarrow$ chart algebra $\mathcal{A}_b \rightarrow$ bar shadow $\overline{B}(\mathcal{A}_b) \rightarrow$ derived chiral centre $Z_{\text{ch}}^{\text{der}}(\mathcal{A}_b)$ (the bulk) $\rightarrow$ acting line operators $\rightarrow$ scalar modular shadow. Identifying stages without the licensing datum -- shadow = object -- is the master error the programme is disciplined against; this volume’s own instrument is the three-axis type signature: level, chart, ambient, declared on every theorem.

The five gardens are five readings of the chain. *Volume I* (chiral-bar-cobar) builds the engine on algebraic curves: bar–cobar adjunction and inversion, quadratic Koszul recognition, centre complementarity, the deformation–Hodge–graph traces $F_g = \kappa \lambda_g^{\text{FP}} + \partial F_g^{\text{cross}}$, Hochschild concentration, and

the arithmetic shadow. *Volume II* (chiral-bar-cobar-vol2) reads the same object as physics: bar = BV/BRST, quantum master equation = Maurer–Cartan, anomaly = curvature, the two-coloured Swiss-cheese operad as bulk–boundary home, Universal Holography $\mathcal{A} \mapsto (\partial = \mathcal{A}, \text{bulk} = Z_{\text{ch}}^{\text{der}}(\mathcal{A}))$, at $\text{Vir}_{c=3\ell/2G_N}$ the boundary reading of 3d gravity. *This volume* constructs the geometric source: the two-stage functor from Calabi–Yau categories, the Hall positive half and its double as the level-three quantum group, and the κ -spectrum as the level-S characteristic. *Mixed holomorphic–topological strings* computes the local germ of the $d = 2$ theory at N Dirac branes -- the trace coordinate $J(f) = \text{Tr } f(\phi_1, \phi_2)$ -- with the holomorphic de Rham obstruction on $K_3 \times E$ showing why the local stalk does not globalise for free. *The Igusa cusp form* (igusa-cusp-form) is the terminal scalar: Δ_5 read five ways, the order-two orientation bit that squaring kills, and the finite recognition data -- this volume’s own frontier -- that would promote the scalar back to the operator $\mathbf{H}_{\Delta_5}$.

The single question underneath: characterise the modular convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ as a geometric object. The proposed answer -- endomorphisms of the Lagrangian embedding $\mathcal{L}_{\mathcal{A}} \hookrightarrow \mathcal{M}_{\text{vac}}(\mathcal{A})$ in a (-2) -shifted symplectic stack of vacua -- has Volume I computing its Taylor jets, Volume II its local composition law, this volume its geometric source, and the satellites its sharpest example: the $K_3 \times E$ datum whose scalar shadow is Δ_5 .

6 The frontier

The volume’s honest split is scalar versus algebraic. Everything scalar and automorphic at $K_3 \times E$ -- the Oberdieck–Pandharipande identity, the Katz–Klemm–Vafa product, the second-quantised elliptic genus, the Borcherds lift and denominator, the η^{-48} Humbert residue, the universal weight $c_N(\mathfrak{o})/2$ -- is proved with sources pinned. The algebraic layer is one coherent conditional programme whose missing inputs are enumerated:

- (i) The compact critical CoHA package for $K_3 \times E$ -- atlas, Serre-compatible quasi-resolution descent, Hopf pairing with Poincaré–Birkhoff–Witt integral form, stable-envelope transport, double descent -- the existence layer of $G(K_3 \times E)$ and $\mathbf{H}_{\Delta_5}$. The deepest form: away from toric charts, most of the global double is invisible to local chart data.
- (ii) The four pentagon hypotheses closing the six-route colimit.
- (iii) The compact CY_3 strictification of stage one: identify $B_{\text{TCFT}}^{(2)}$ with $B_{\text{term}}^{(2)}$ at chain level; until then the quintic carries curved formality with Yukawa curving.
- (iv) Chain fusion in general dimension -- the bridge from this volume’s stage-two output to the open-side primitive package of Volumes I and II.
- (v) The non-abelian K_3 Yangian (lattice-Hopf matching, Jacobi at the vertex level); the Fake Monster at $d = 5$; the $\mathcal{W}_{\infty}[\lambda]$ endpoint beyond its admissible window; modularity under fusion.

The volume’s architectural invention matches the Igusa companion’s: recognition criteria with finite data lists, three-axis scope signatures, and the discipline that a character identity is never promoted to an algebra identification. The geometric source is constructed exactly as far as the obstructions permit, and the obstructions are theorems.